

Exercise 3: Assertions in JUnit

Calculator.java:

```
package com.exercise;

public class Calculator {
    public int add(int a, int b) {
        return a + b;
    }
}
```

CalculatorTest.java:

```
package com.exercise;

import static org.junit.Assert.assertEquals;
import org.junit.Test;

public class CalculatorTest {
    @Test
    public void testAdd() {
        Calculator calc = new Calculator();
        assertEquals(5, calc.add(2, 3));
    }
}
```

AssertionsTest.java:

```
package com.example;

import static org.junit.Assert.assertEquals;
import static org.junit.Assert.assertFalse;
import static org.junit.Assert.assertNotNull;
import static org.junit.Assert.assertNull;
import static org.junit.Assert.assertTrue;
import org.junit.Test;

public class AssertionsTest {
```

```

@Test
public void testAssertions() {

    assertEquals(5, 2 + 3);

    assertTrue(5 > 3);

    assertFalse(5 < 3);

    assertNull(null);

    assertNotNull(new Object());

}
}

```

Output:

```

adityahargunani@Adityas-MacBook demo % mvn test -Dtest=AssertionsTest
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:demo >-----
[INFO] Building demo 1.4
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.0.2:resources (default-resources) @ demo ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory /Users/adityahargunani/Downloads/Exercise 3- Assertions in JUnit/demo/src/main/resources
[INFO]
[INFO] --- compiler:3.8.0:compile (default-compile) @ demo ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 1 source file to /Users/adityahargunani/Downloads/Exercise 3- Assertions in JUnit/demo/target/classes
[INFO]
[INFO] --- resources:3.0.2:testResources (default-testResources) @ demo ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory /Users/adityahargunani/Downloads/Exercise 3- Assertions in JUnit/demo/src/test/resources
[INFO]
[INFO] --- compiler:3.8.0:testCompile (default-testCompile) @ demo ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 2 source files to /Users/adityahargunani/Downloads/Exercise 3- Assertions in JUnit/demo/target/test-classes
[INFO]
[INFO] --- surefire:2.22.1:test (default-test) @ demo ---
[INFO]
[INFO] T E S T S
[INFO]
[INFO] Running com.example.AssertionsTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.018 s - in com.example.AssertionsTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.124 s
[INFO] Finished at: 2026-06-23T15:55:47+05:30
[INFO]
adityahargunani@Adityas-MacBook demo %

```

